

Diatonicism vs Chromaticism: data-driven perspectives on style in music

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Abstract: Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Keywords: 1–5 keywords.

I. INTRODUCTION

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*I am grateful to Robert Lieck and Zeng Ren for valuable discussions.

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Text requiring further explanation¹.

II. METHOD

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i. Data

Use *Enhanced Wikifonia Leadsheet Dataset*² (EWLD)? Compare with other historical corpora (e.g. Polish scores corpus, DCML data, CRIM).

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ii. Different scale orderings

Fifth vs chromatic ordering. Well-formed scales

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¹Example footnote

²https://zenodo.org/record/1476555#.ZP_9TdJBxhF

Table 1: *Datasets used in the subsequent analyses.*

Name	Styles	Extent	Link
Tonal Pitch-Class Counts Corpus (TP3C)	Western Classical	1300–2000	[4]
Classical Archives Corpus	Pop/Rock		
Lakh MIDI	Pop		
Enhanced Wikifonia Leadsheet Dataset	Pop/Rock		

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iii. DFT

DFT-products [5]

DFT matrix for 12 pitch classes

Adapted from Wikipedia:

An N -point DFT is expressed as the multiplication $X = Wx$, where x is the original input signal, W is the N -by- N square DFT matrix, and X is the DFT of the signal.

The transformation matrix W can be defined as:

$$W = \left(\frac{\omega^{jk}}{\sqrt{N}} \right)_{j,k=0,\dots,N-1} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \dots & \omega^{2(N-1)} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \dots & \omega^{3(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \omega^{3(N-1)} & \dots & \omega^{(N-1)(N-1)} \end{bmatrix}, \quad (1)$$

where $\omega = e^{-2\pi i/N}$ is a primitive N th root of unity in which $i^2 = -1$. We can avoid writing large exponents for ω using the fact that for any exponent x we have the identity

$$\omega^x = \omega^{x \bmod N}. \quad (2)$$

This is the Vandermonde matrix for the roots of unity, up to the normalization factor. Note that the normalization factor in front of the sum ($1/\sqrt{N}$) and the sign of the exponent in ω are merely conventions, and differ in some treatments. All of the following discussion applies regardless of the convention, with at most minor adjustments. The only important thing is that the forward and inverse transforms have opposite-sign exponents, and that the product of their normalization factors be $1/N$. However, the $1/\sqrt{N}$ choice here makes the resulting DFT matrix unitary, which is convenient in many circumstances.

Fast Fourier transform algorithms utilize the symmetries of the matrix to reduce the time of multiplying a vector by this matrix, from the usual $O(N^2)$. Similar techniques can be applied for multiplications by matrices such as Hadamard matrix and the Walsh matrix.

iii.1 Pasted from ChatGPT: A Fourier-Theoretic Measure of Structural Preservation

Let P be a probability distribution over the twelve pitch classes, viewed as a function on the cyclic group

$$\mathbb{Z}_{12} = \{0, \dots, 11\},$$

with

$$P(x) \geq 0, \quad \sum_{x \in \mathbb{Z}_{12}} P(x) = 1.$$

Rather than viewing P simply as a vector of probabilities, we regard it as a function on a finite Abelian group. This allows us to analyze its structure using harmonic analysis, where the natural basis is given by the irreducible characters of $\mathbb{Z}_{12}$.

The discrete Fourier transform (DFT) of P is

$$\hat{P}(k) = \sum_{x=0}^{11} P(x) e^{-2\pi i k x / 12}, \quad k = 0, \dots, 11.$$

The inverse transform is

$$P(x) = \frac{1}{12} \sum_{k=0}^{11} \hat{P}(k) e^{2\pi i k x / 12}.$$

Thus the Fourier coefficients provide an alternative coordinate system for describing the pitch-class distribution.

iv. Interpretation of the Fourier coefficients

Each Fourier coefficient corresponds to a periodic component of the pitch-class distribution.

The zeroth coefficient,

$$\hat{P}(0) = 1,$$

contains only the total probability mass.

The remaining coefficients describe cyclic periodicities.

For example,

k	Approximate musical interpretation
1	global orientation of the chromatic circle
2	whole-tone polarity
3	minor-third / diminished-seventh periodicity
4	augmented-triad periodicity
6	tritone polarity

Since P is real-valued,

$$\hat{P}(12 - k) = \overline{\hat{P}(k)},$$

so only six independent complex coefficients exist.

The magnitudes

$$|\hat{P}(k)|$$

measure the strength of each periodicity, while the phases specify where that periodicity is located on the chromatic circle.

This decomposition is well established in computational music theory, where the Fourier coefficients are often interpreted as descriptors of intervallic structure.

v. Automorphisms of the chromatic circle

The multiplicative units of $\mathbb{Z}_{12}$ are

$$\{1, 5, 7, 11\},$$

each defining an automorphism of the cyclic group.

Here we consider

$$\pi(x) = 7x \pmod{12}.$$

Since

$$7^2 \equiv 1 \pmod{12},$$

the map is an involution.

If

$$Q(x) = P(\pi^{-1}(x)),$$

then because $7^{-1} = 7$,

$$Q(x) = P(7x).$$

This operation merely reorders the pitch classes.

Importantly, Shannon entropy, Kullback–Leibler divergence, and Jensen–Shannon divergence cannot distinguish P from Q , since they depend only on the multiset of probabilities rather than on their assignment to pitch classes.

The Fourier transform, however, reveals how the automorphism acts on musical structure.

vi. Action on the Fourier spectrum

The Fourier transform of Q is

$$\Phi(k) = \sum_x Q(x) e^{-2\pi i k x / 12}.$$

Substituting $Q(x) = P(7x)$ and changing variables gives

$$\Phi(k) = \Phi(7k \bmod 12).$$

Thus the automorphism does not distort the Fourier coefficients.

Instead, it permutes them.

The induced permutation is

k	0	1	2	3	4	5	6	7	8	9	10	11
$7k \bmod 12$	0	7	2	9	4	11	6	1	8	3	10	5

Hence

$$\begin{aligned} 0 &\leftrightarrow 0, \\ 2 &\leftrightarrow 2, \\ 4 &\leftrightarrow 4, \\ 6 &\leftrightarrow 6, \\ 8 &\leftrightarrow 8, \\ 10 &\leftrightarrow 10, \end{aligned}$$

while

$$(1\ 7), \quad (3\ 9), \quad (5\ 11)$$

are exchanged.

Consequently, all even Fourier modes are invariant under multiplication by seven, whereas the odd modes are reflected.

This decomposition provides a natural distinction between structural features that are preserved by the automorphism and those that are transformed.

vii. Spectral energy

By Parseval's identity,

$$\sum_{x=0}^{11} |P(x)|^2 = \frac{1}{12} \sum_{k=0}^{11} |\mathfrak{p}(k)|^2.$$

The quantities

$$|\mathfrak{p}(k)|^2$$

may therefore be interpreted as the distribution of structural energy across Fourier modes. Normalizing gives

$$E_k = \frac{|\mathfrak{p}(k)|^2}{\sum_{j=0}^{11} |\mathfrak{p}(j)|^2},$$

which satisfies

$$E_k \geq 0, \quad \sum_k E_k = 1.$$

Thus $\{E_k\}$ forms a probability distribution over harmonic components rather than over pitch classes.

viii. Information retained by the automorphism

Because multiplication by seven leaves the even Fourier modes fixed, a natural measure of structural preservation is the proportion of spectral energy contained within those invariant modes,

$$R = \frac{\sum_{k \in \{0,2,4,6,8,10\}} |\mathfrak{p}(k)|^2}{\sum_{k=0}^{11} |\mathfrak{p}(k)|^2}.$$

The quantity R lies between 0 and 1.

Values near one indicate that most of the harmonic structure resides in Fourier modes that are invariant under the automorphism, implying that multiplication by seven preserves most of the distribution's structural organization.

Conversely, values near zero indicate that the structure depends primarily on odd Fourier modes, which are exchanged by the automorphism.

A more refined measure compares the entire spectra before and after transformation,

$$C = \sum_{k=0}^{11} w_k \mathfrak{P}(k) \overline{\mathfrak{P}(7k)},$$

where w_k are non-negative weights reflecting the relative importance of different periodicities. Normalizing yields

$$\rho = \frac{\sum_k w_k \Re(\mathfrak{P}(k) \overline{\mathfrak{P}(7k)})}{\sum_k w_k |\mathfrak{P}(k)|^2},$$

which may be interpreted as a spectral correlation coefficient.

Unlike Shannon-theoretic measures, which are invariant under any permutation of the underlying probabilities, these Fourier-based quantities are sensitive to the algebraic action of the automorphism on the harmonic structure of the pitch-class distribution.

They therefore quantify not whether information is preserved in the information-theoretic sense, but rather how much of the musically meaningful periodic organization remains invariant under the transformation.

ix. Pitch-class distributions as areas

We define the *area* given by a pc distribution as

$$A = \sum_{i,j=0}^{11} \frac{p_i p_j}{d(i, j)}, \quad (3)$$

where p_i is the the relative frequency of pitch class i in some musical context (measure, segment, piece) and $d(i, j)$ is the *distance* between them.

Several distance metrics are possible...

x. Probability distributions/sampling from pieces

Do not only plot single PCVs/PCDs in StarPlots, but also superimpose several samples from the same piece, e.g. using DimCAT or keyscapes...

III. ANALYSES OF CHARACTERISTIC PIECES

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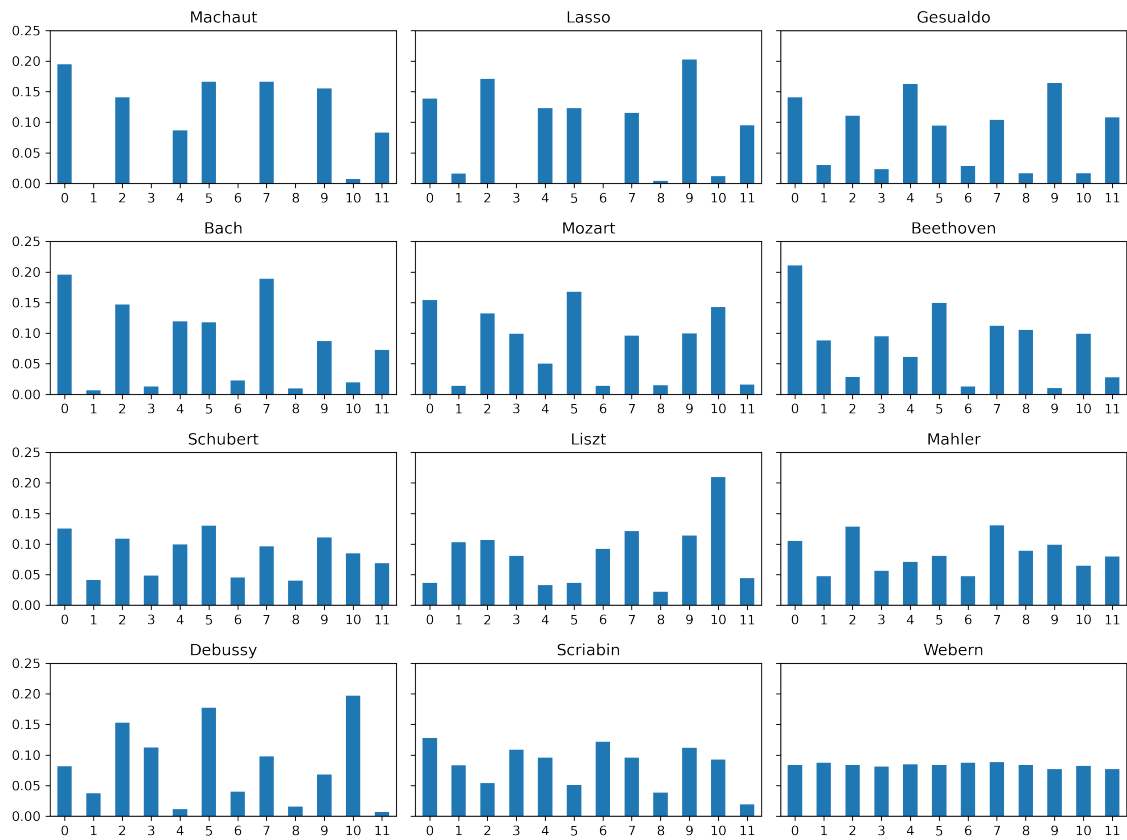


Figure 1: Comparison of pitch-class distributions between pieces by composers from different historical periods.

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i. Modal music

Maybe instead of analyzing all individual pieces here, focus rather on modes?

i.1 Machaut

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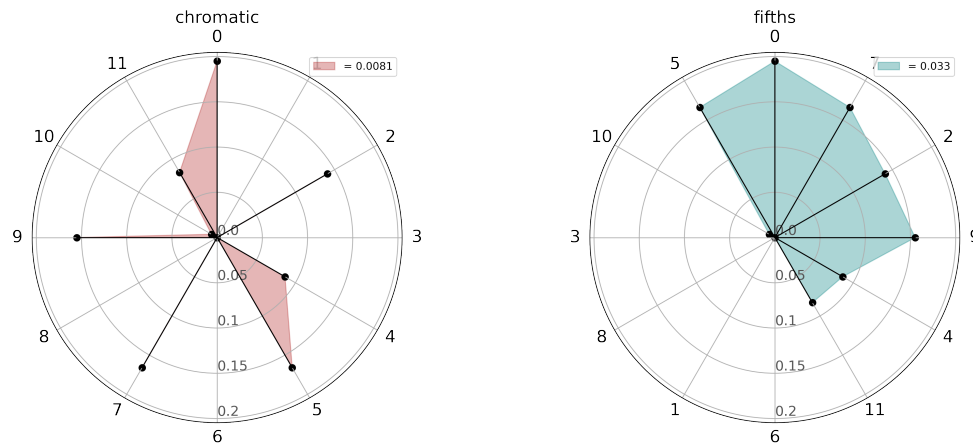


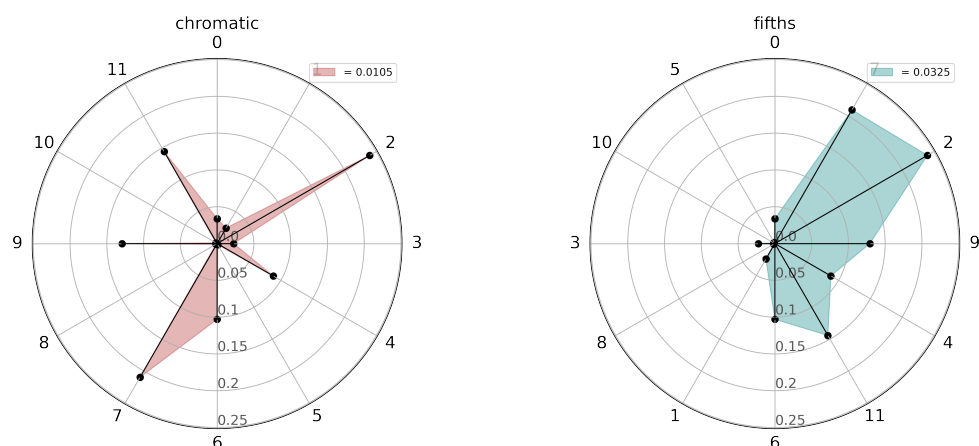
Figure 2: Machaut

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i.2 Gesualdo

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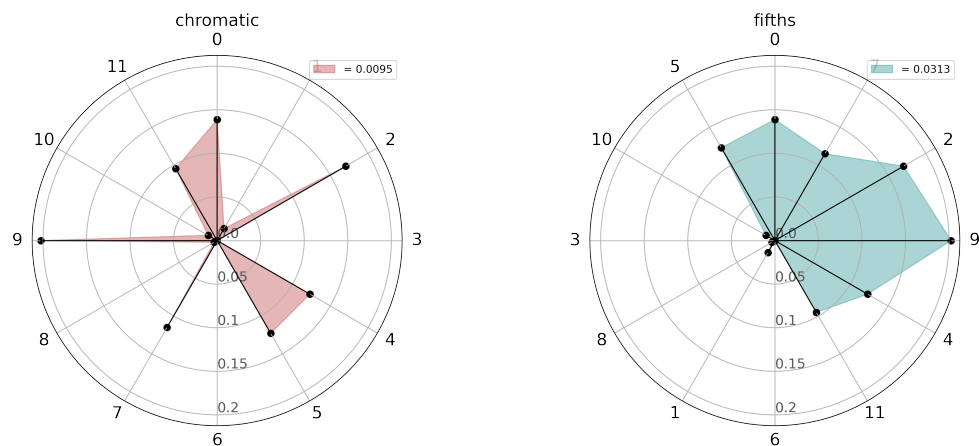
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Figure 3: *Gesualdo*

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i.3 Lasso

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Figure 4: *Lasso*

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ii. Diatonicism results in asymmetrical pitch-class distributions

Instead of great composers, compare major vs minor? OR, rather: focus on the diatonicism vs chromaticism angle (see also [3]). Point is that there is a high correlation between historical period and "syntax", but there are exceptions (Gesualdo, Debussy)

ii.1 Debussy

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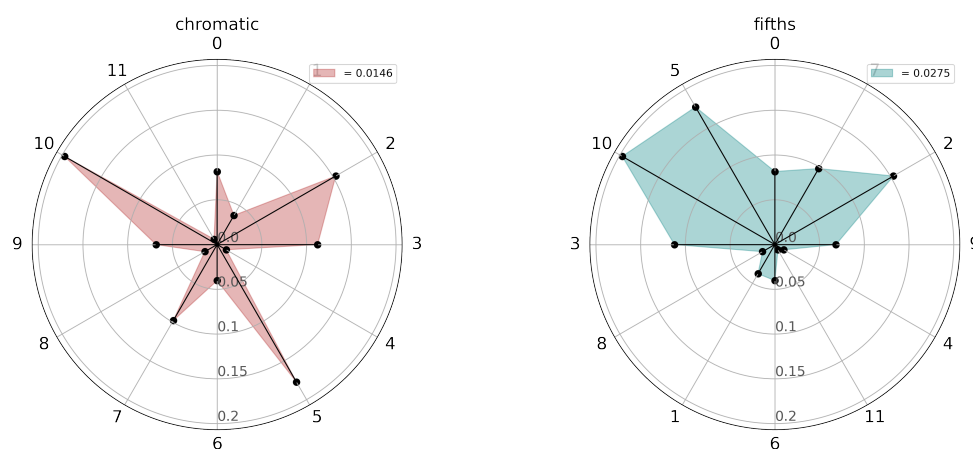


Figure 5: Debussy

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ii.2 Bach

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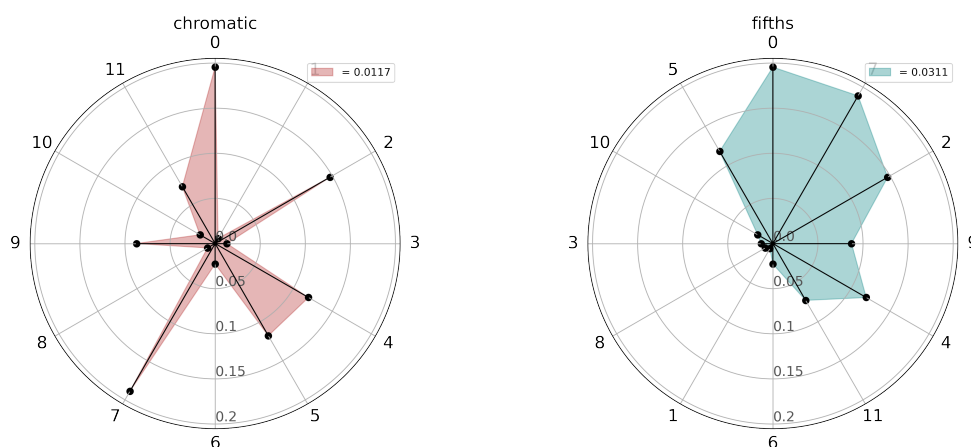


Figure 6: Bach

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ii.3 Mozart

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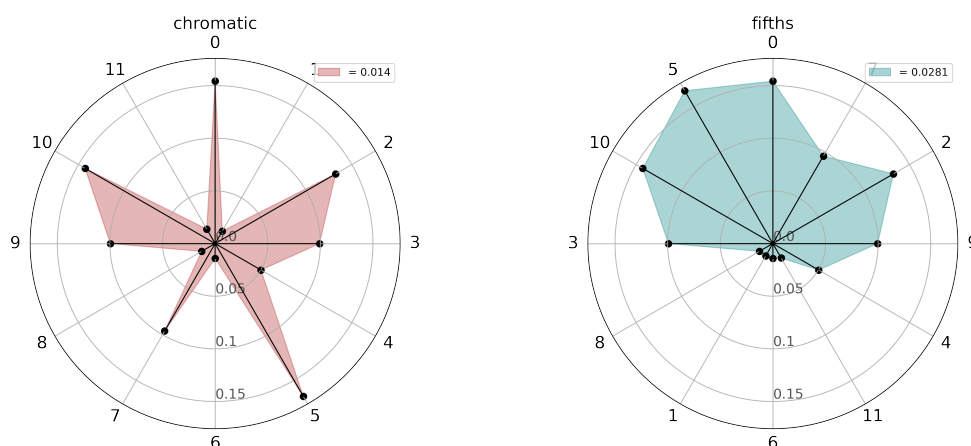


Figure 7: Mozart

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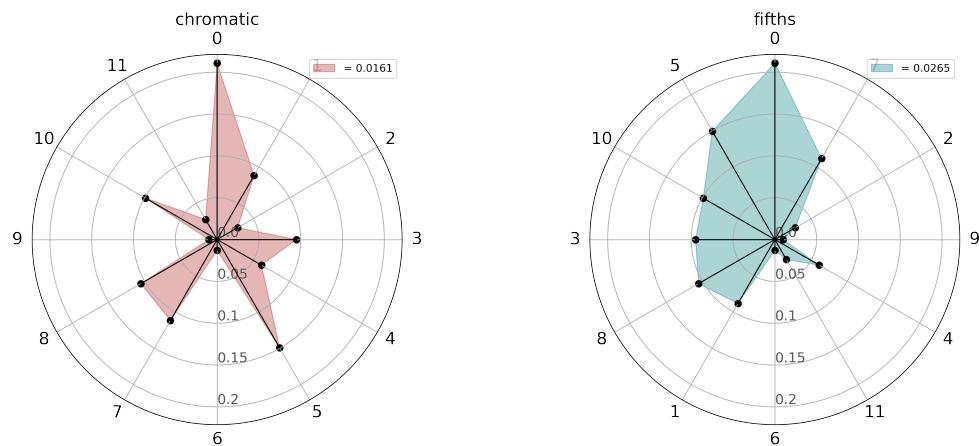
ii.4 Beethoven

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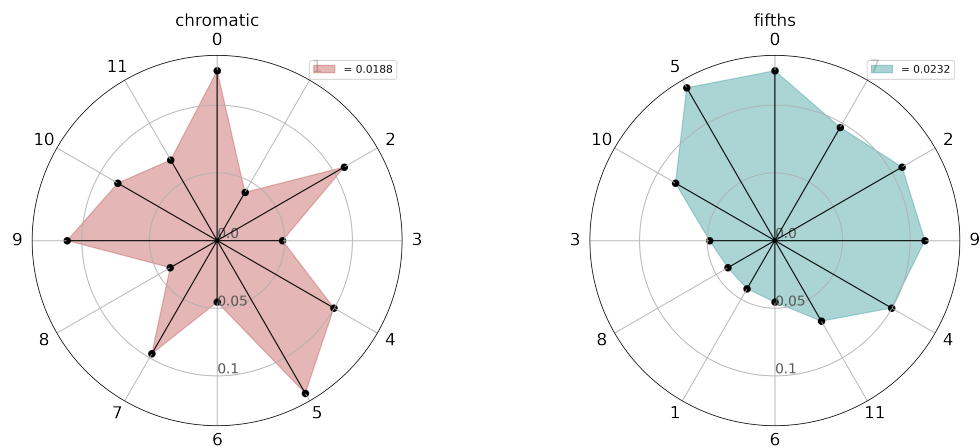
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ii.5 Schubert

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Figure 8: *Beethoven*

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Figure 9: *Schubert*

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ii.6 Hensel

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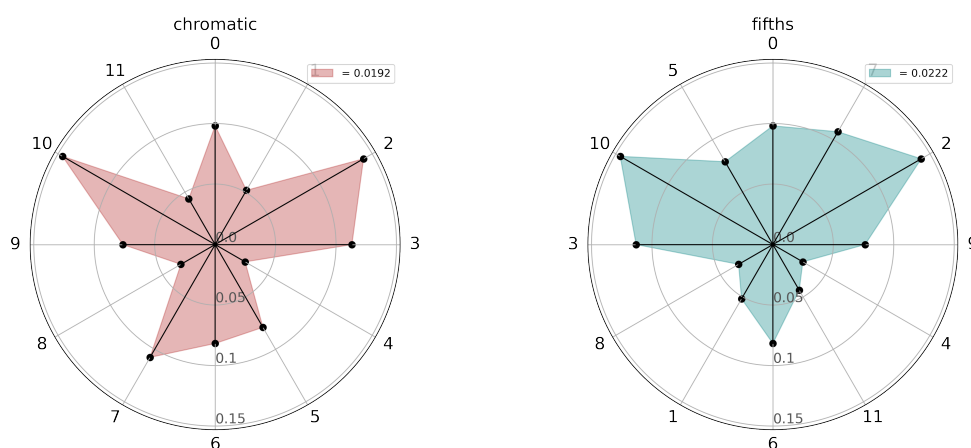


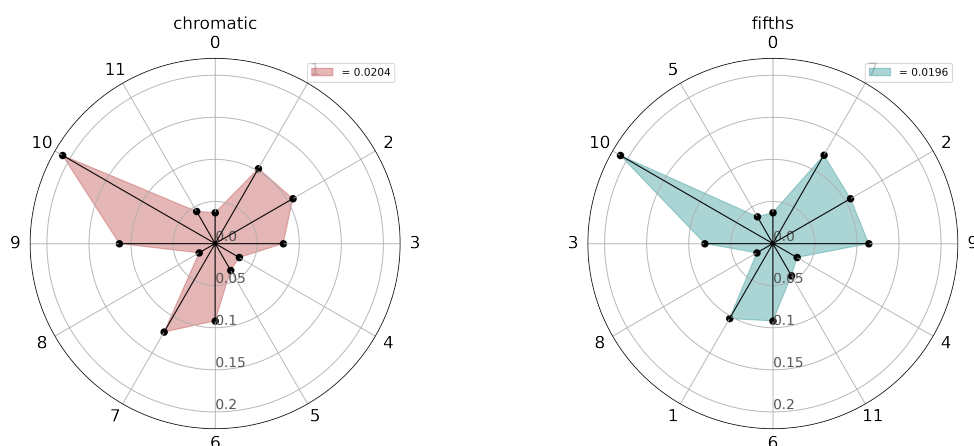
Figure 10: Hensel

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iii. Chromaticism features nearly-symmetrical pitch-class distributions

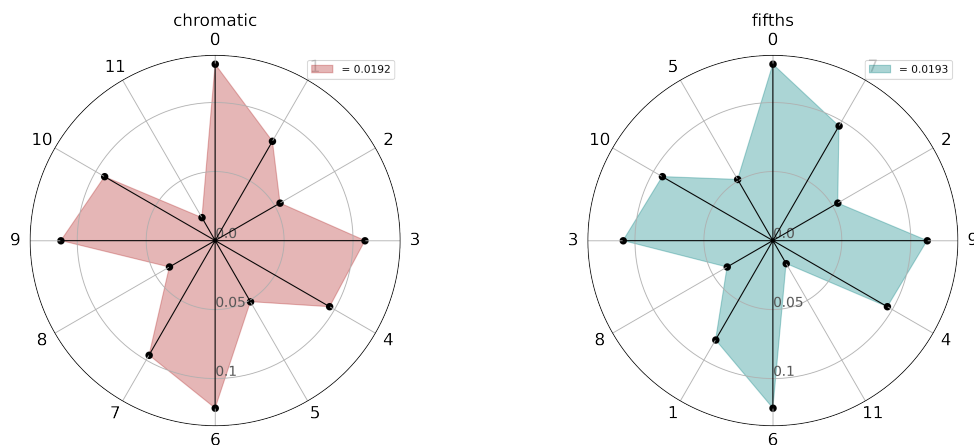
iii.1 Liszt

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Figure 11: *Webern*

iii.2 Scriabin

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Figure 12: *Scriabin*

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iii.3 Mahler

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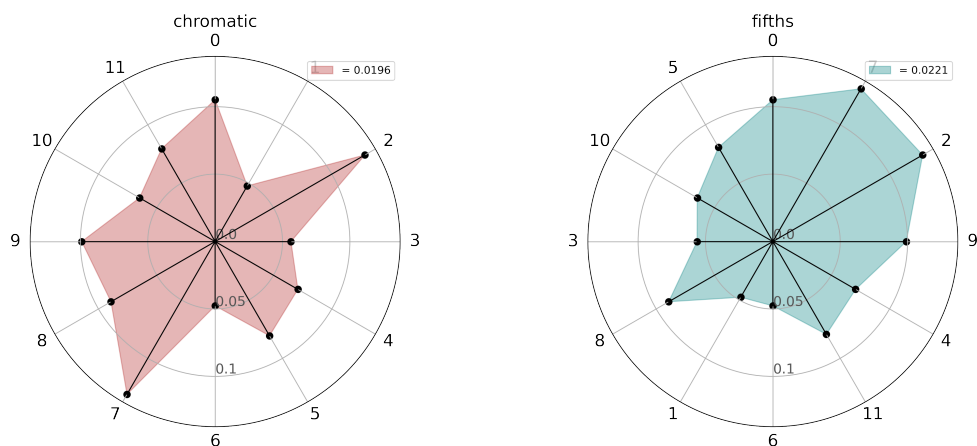


Figure 13: Mahler

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iii.4 Vaughan-Williams

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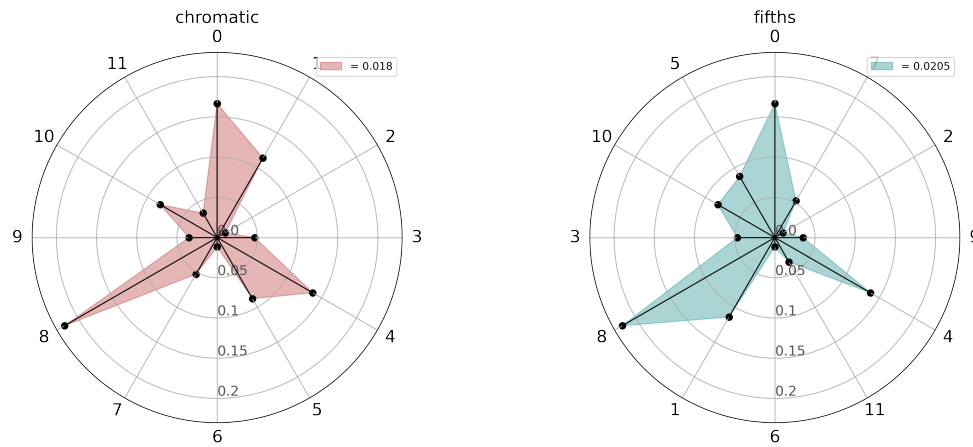


Figure 14: Vaughan-Williams

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iii.5 Webern

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See example in Table 1.

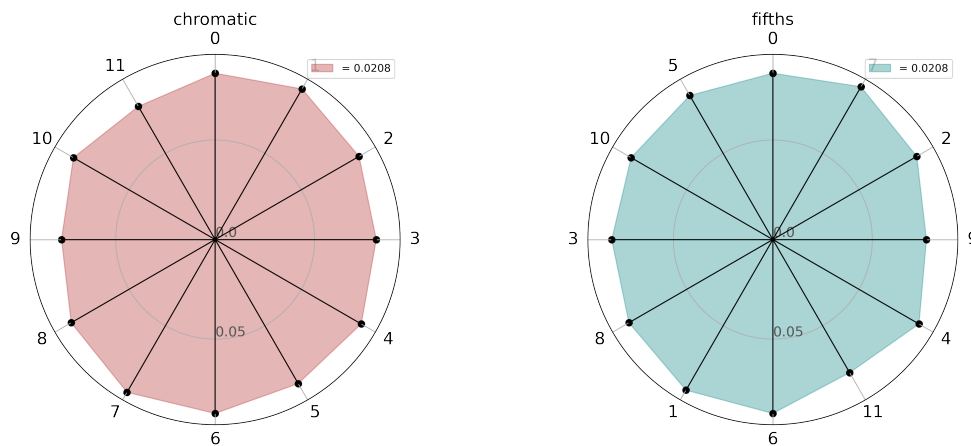


Figure 15: Webern

IV. HIERARCHICAL ANALYSIS

Wavescapes, pitchscapes, samples to StarPlots

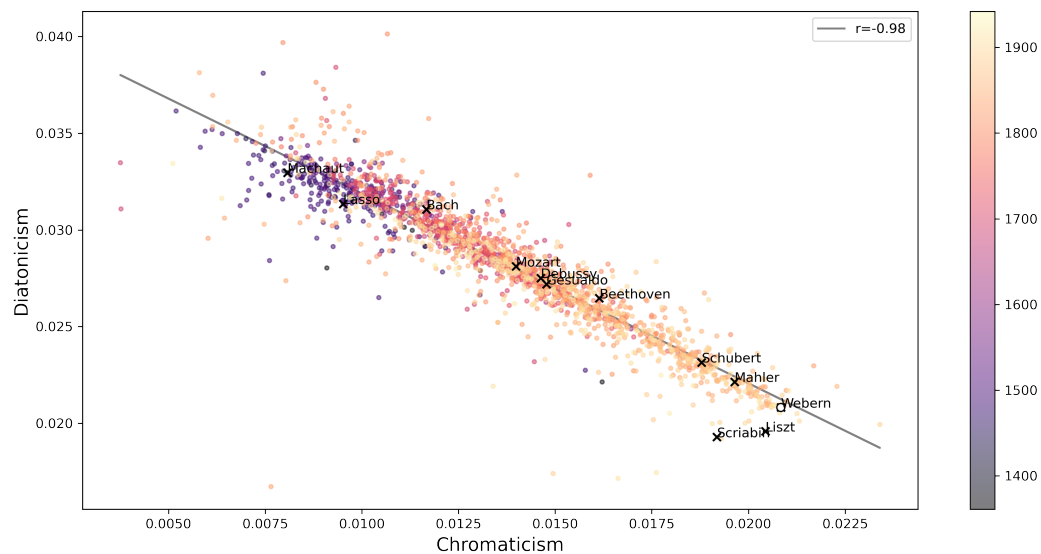
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V. DIACHRONIC ANALYSES

- i. Diatonicism vs Chromaticism
- ii. Comparing different styles and genres

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Important: add also composers' names that do *not* fit the narrative. Discuss here briefly, and in more detail in the general discussions that example-based music historiography and style analysis is prone to confirmation bias regarding a linear, progressive version of music ("evolution" in the wrong sense). It is particularly important to also regard conflicting examples. It is crucial to determine whether those are 'outliers', or whether they refute one's theory of style

Figure 16: *Diatonicism vs Chromaticism***evolution.**

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ii.1 Western classical music

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ii.2 Pop/Rock

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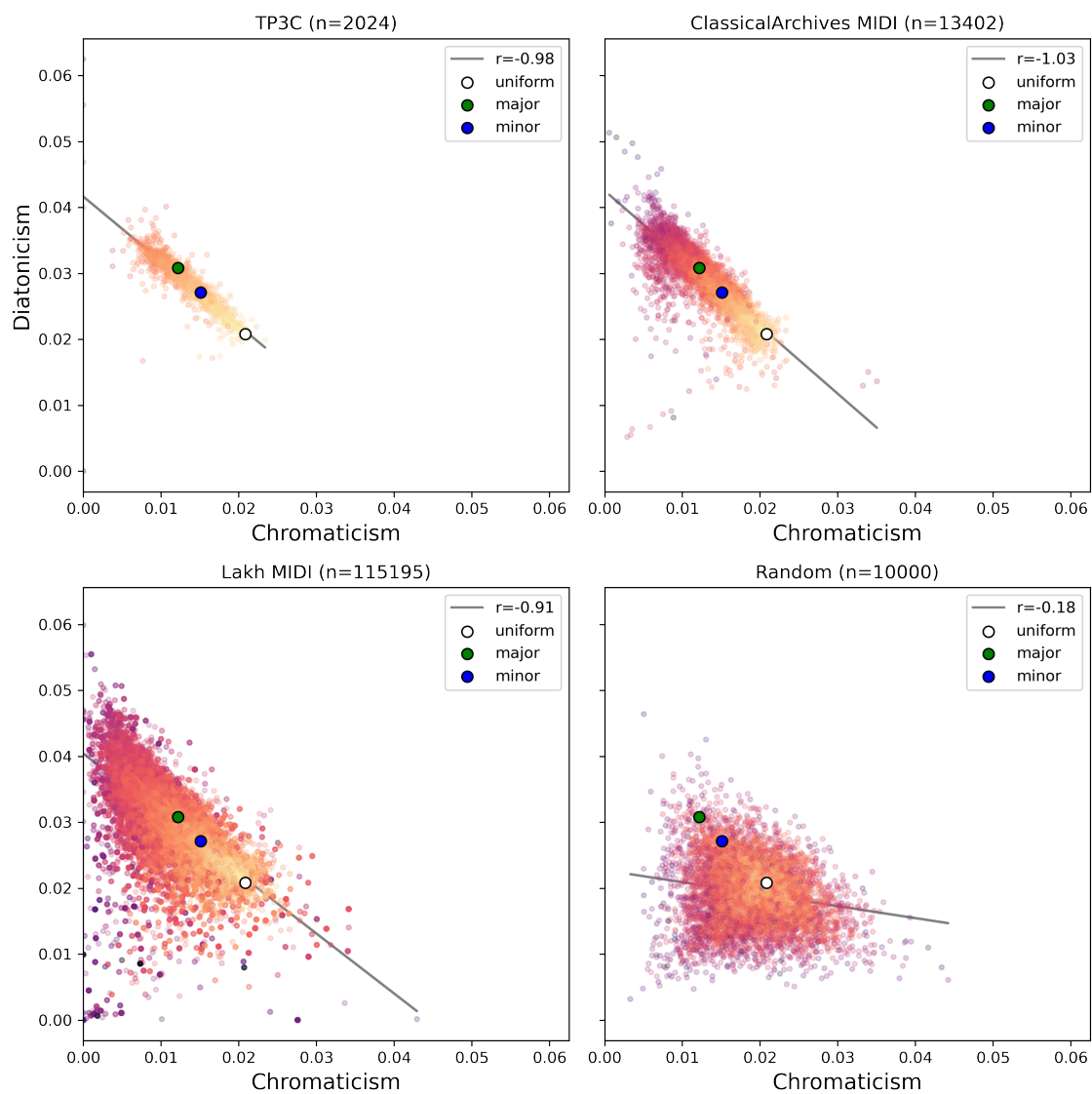


Figure 17: Comparison of large-scale behavior in different datasets.

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VI. GENERAL DISCUSSION & CONCLUSIONS

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$$e = mc^2 \quad (4)$$

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